What is Networks

One efficient way of solving complex problems is following the lemma “divide and conquer”.

A complex system may be decomposed into simpler elements, in order to be able to understand it. Also simple elements may be gathered to produce a complex system. Networks are one approach for achieving this.

There are a large number of different types of networks, but they all are characterized by the following components: a set of nodes, and connections between nodes. The nodes can be seen as computational units. They receive inputs, and process them to obtain an output.

What is Networks

This processing might be very simple (such as summing the inputs), or quite complex (a node might contain another network...) the connections determine the information flow between nodes.

They can be unidirectional, when the information flows only in one sense, and bidirectional, when the information flows in either sense.

Networks are used to model a wide range of phenomena in physics, computer science, biochemistry, ethology, mathematics, sociology, economics, telecommunications, and many other areas.

This is because many systems can be seen as a network: proteins, computers, communities, etc.

What is Neural Network

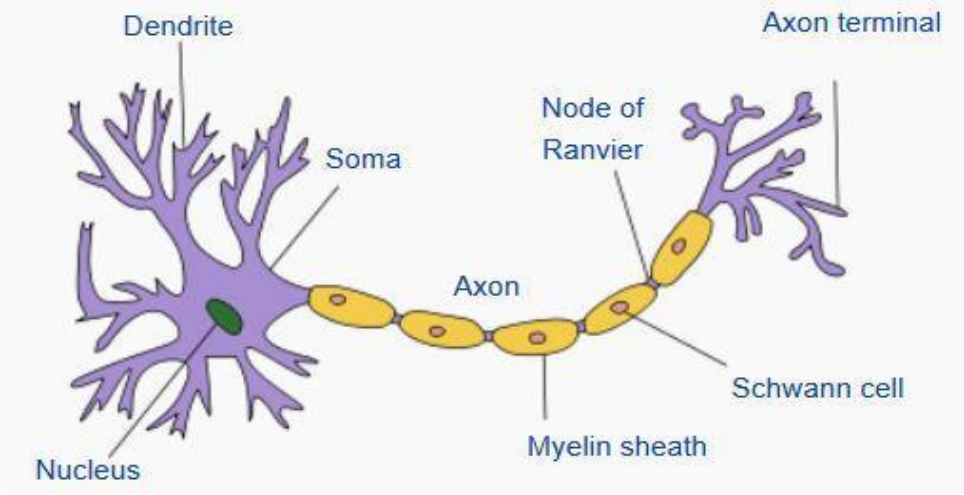
A **neural network** is a system of **interconnected neurons** that processes input data by passing it through layers and producing an output — typically used in **machine learning** to solve complex tasks like classification, prediction, and pattern recognition.

#Example:

- Classify whether an email is **spam** or **not spam** **(**Classification**)**
- Predict tomorrow's **temperature (**Prediction**)**
- Recognize handwritten digits (like 0–9) (Pattern Recognition)

Biological Neurons

Structure of Biological Neuron



Biological Neuron

A biological neuron has three types of components that are of particular interest in understanding an artificial neuron:

1. Dendrites
2. Soma
3. Axon

Biological Neuron - Dendrites

- Receive incoming signals from other neurons. The signals are electric impulses that are transmitted across a synaptic gap by means of a chemical process.
- The action of the chemical transmitter modifies the incoming signal (typically, by scaling the frequency of the signals that are received) in a manner similar to the action of the weights in an artificial neural network.

Biological Neuron - Soma

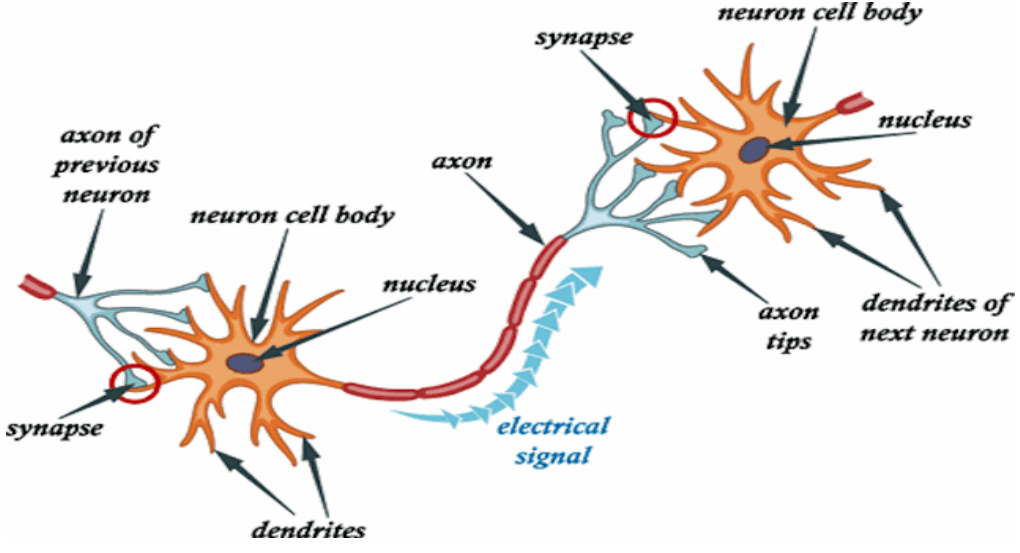
- The soma, or cell body, is responsible for processing the input (incoming signals) and carries biochemical information.
- When sufficient input is received, the cell fires; that is, it transmits a signal over its axon to other cells. It is often supposed that a cell either fires or doesn't at any instant of time, so that transmitted signals can be treated as binary. However, the frequency of firing varies and can be viewed as a signal of either greater or lesser magnitude. This corresponds to looking at discrete time steps and summing all activity (signals received or signals sent) at a particular point in time.

Biological Neuron - Axon

- Responsible for the transmission of signals.
- The transmission of the signal from a particular neuron is accomplished by an action potential resulting from differential concentrations of ions on either side of the neuron's axon sheath (the brain's "white matter"). The ions most directly involved are potassium, sodium, and chloride.

Synapse is present at the end of the axon and is responsible for connecting other neurons.

Biological Neuron



The Neuron

Sensory neurons, **motor neurons** and **interneurons** form the **human brain**. Artificial neurons, form the replica of the human brain (i.e. a neural network). Neural networks are a set of algorithms that are modelled after the human brain.

Artificial Neural Networks

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Artificial Neural Networks (ANN)

- Artificial Neural Network (ANN) is a deep learning algorithm that emerged and evolved from the idea of **Biological Neural Networks of human brains**. An attempt to simulate the workings of the human brain culminated in the emergence of ANN. ANN works very similar to the biological neural networks but doesn't exactly resemble its workings.

- ANN algorithm would accept only **numeric and structured data** as input. To accept **unstructured and non-numeric data** formats such as Image, Text, and Speech, **Convolutional Neural Networks (CNN),** and **Recurrent Neural Networks (RNN)** are used respectively.

Artificial Neural Network (ANN)

- ANN is a **general term** that includes various types of neural networks. In some ANNs like **Recurrent Neural Networks (RNNs)**, **Feedback Neural Networks etc.**, data can flow **in both directions** (forward and backward), depending on the type of network. These networks can have **cycles or loops**, meaning output from later layers may influence earlier layers.

Artificial Neural Network (ANN)

ANN resembles the brain in two respects:

1. Knowledge is acquired by the network from its environment through a learning process.
2. Interneuron connection strengths, known as synaptic weights, are used to store the acquired knowledge.

A neural net consists of a large number of simple processing elements called neurons, units, cells, or nodes. Each neuron is connected to other neurons by means of directed communication links, each with an associated weight. The weights represent information being used by the net to solve a problem.

The basic unit of Artificial Neural Networks

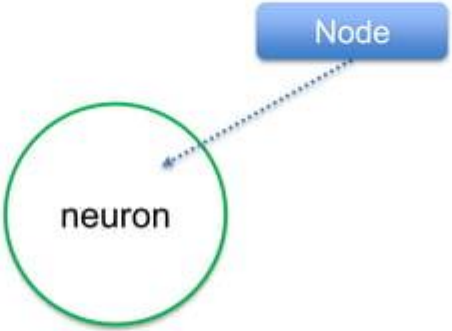
The basic unit of neural networks, the artificial neurons, simulates the four basic functions of natural neurons (receives inputs from other sources, combines them in some way, performs a generally nonlinear operation on the result, and then output the final result).

Here we identify four basic elements of the artificial neural model:

1. Inputs
2. Sum Function,
3. Activation Function and
4. Outputs

Artificial neuron

The Neuron



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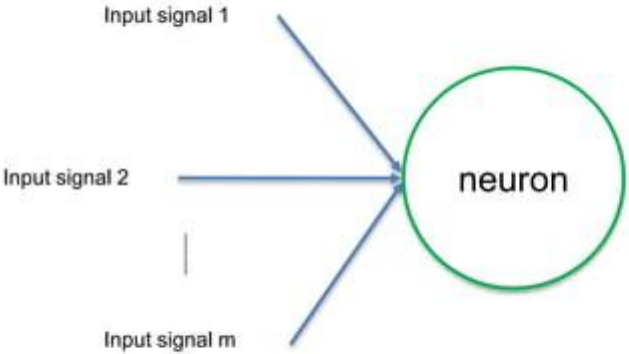

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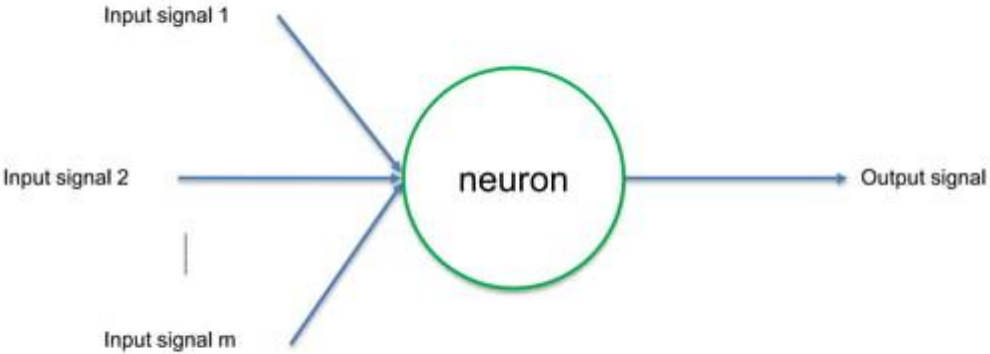

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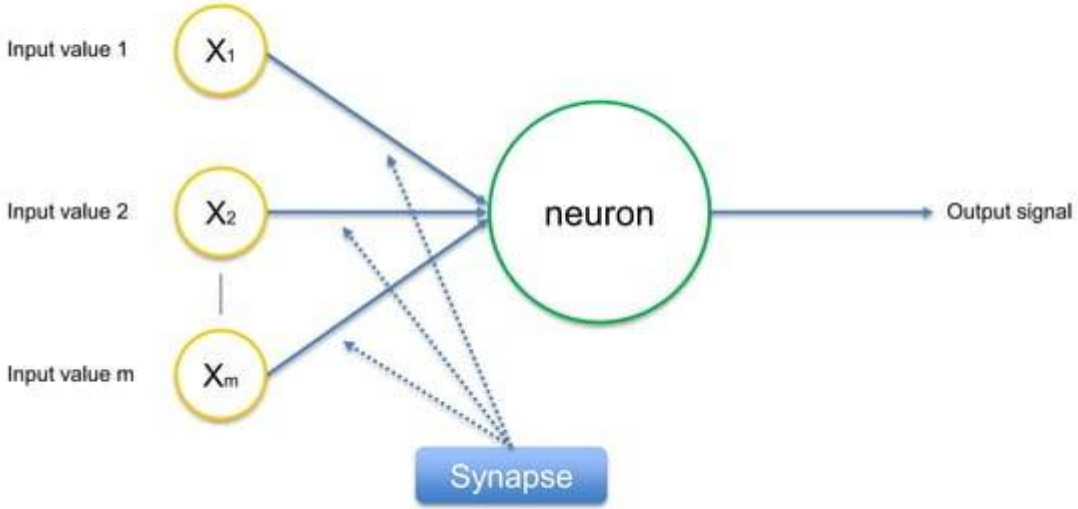

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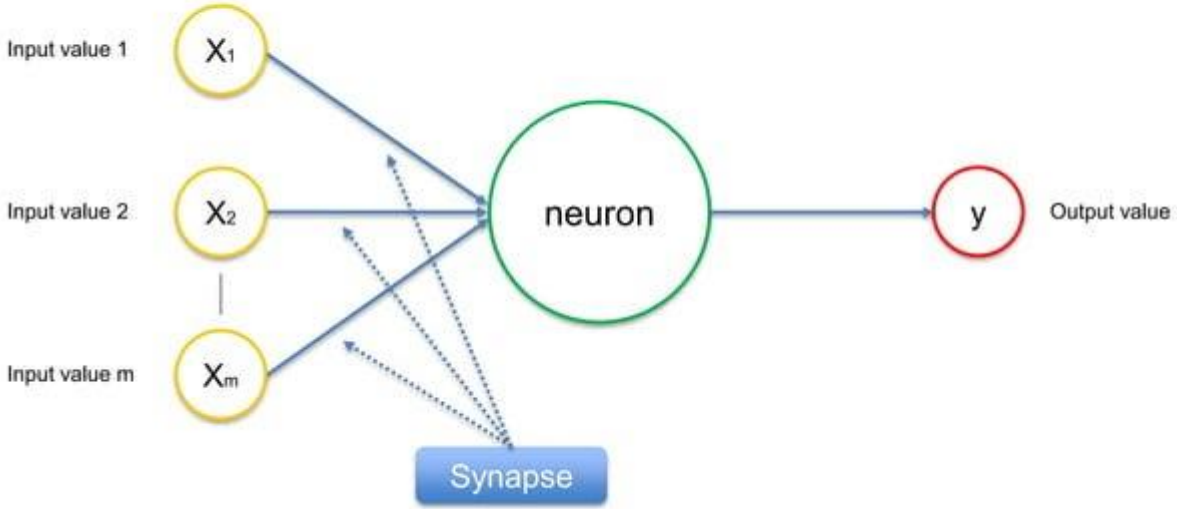
The Neuron



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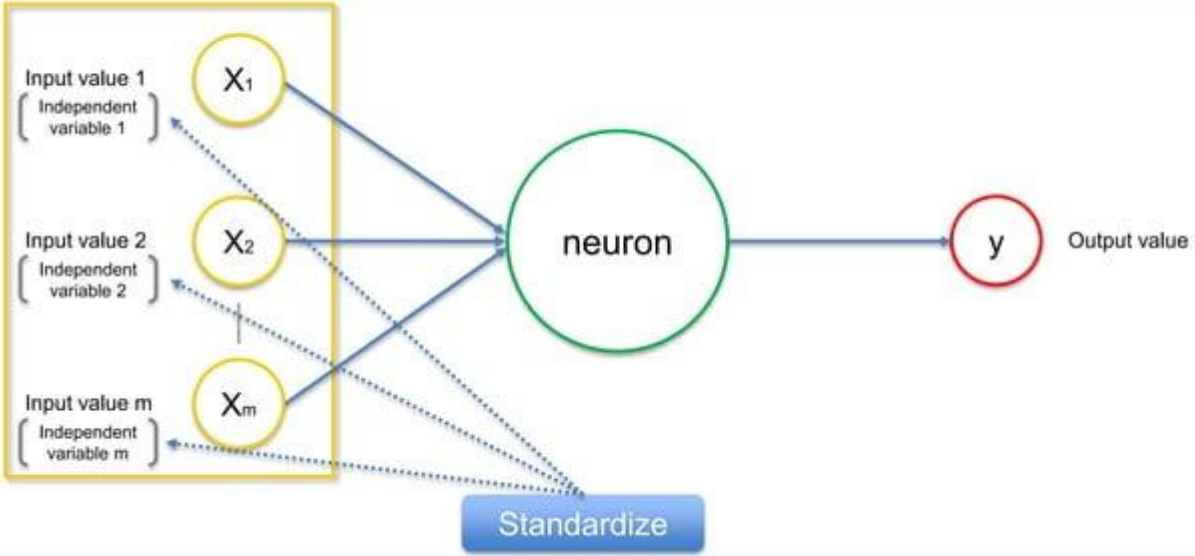

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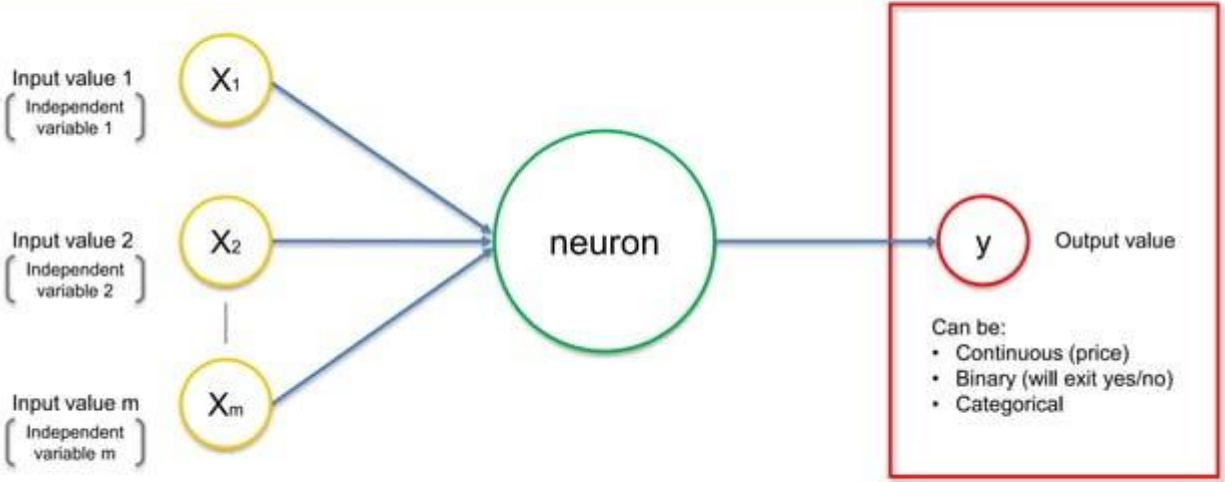

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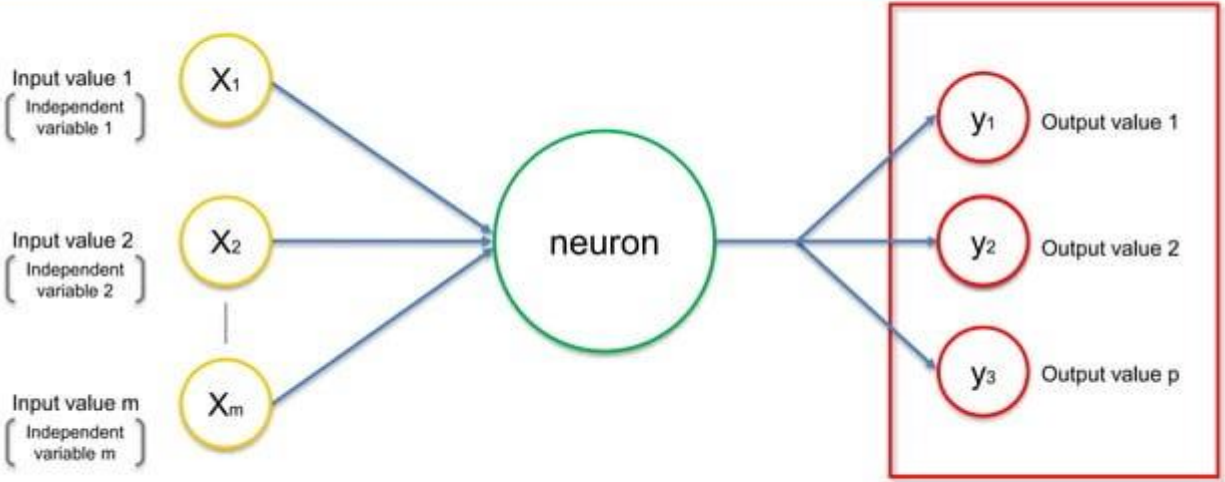

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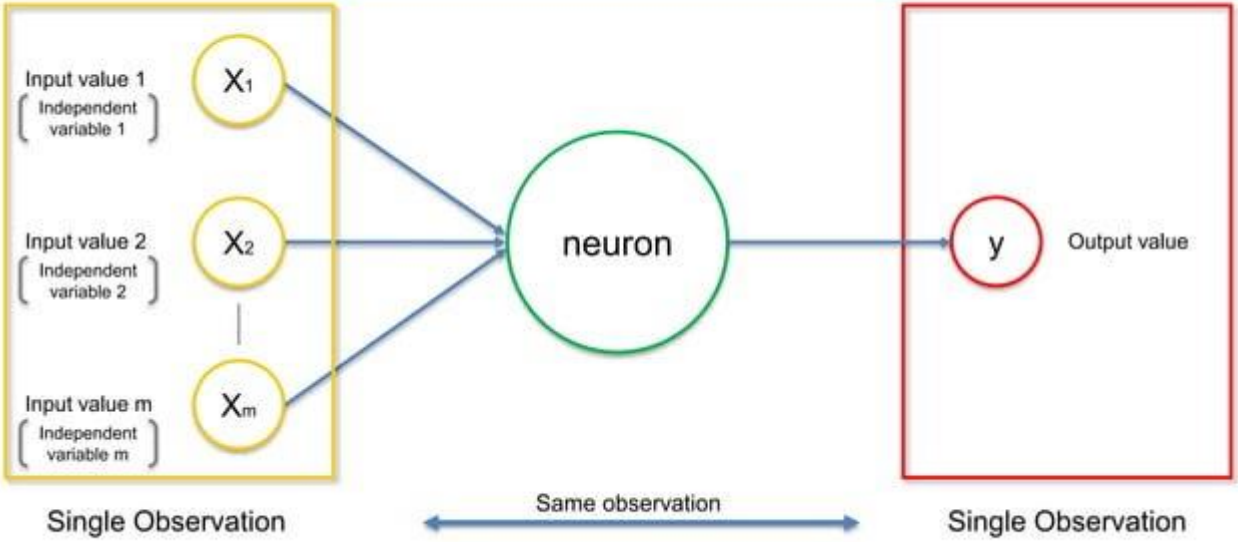

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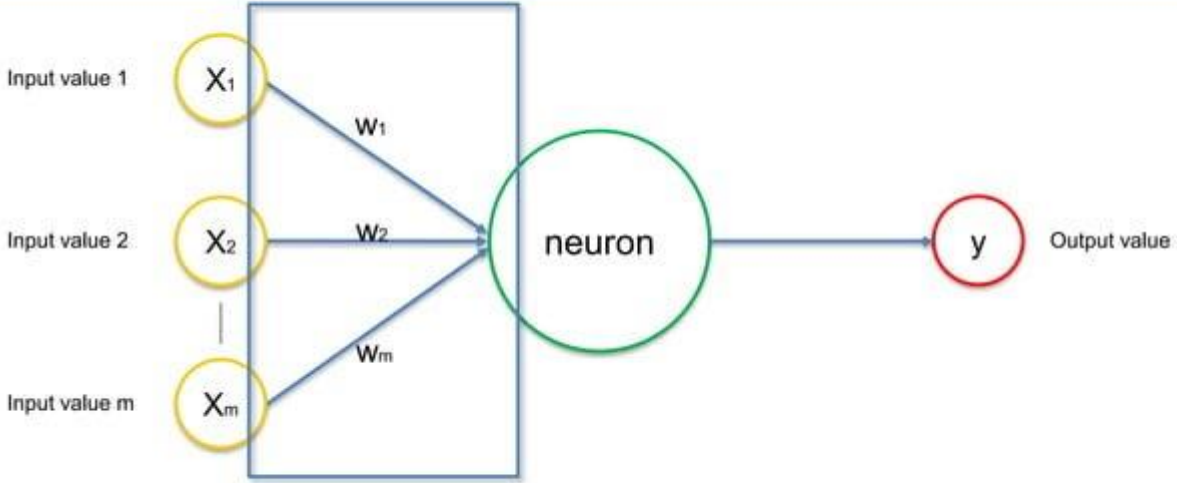

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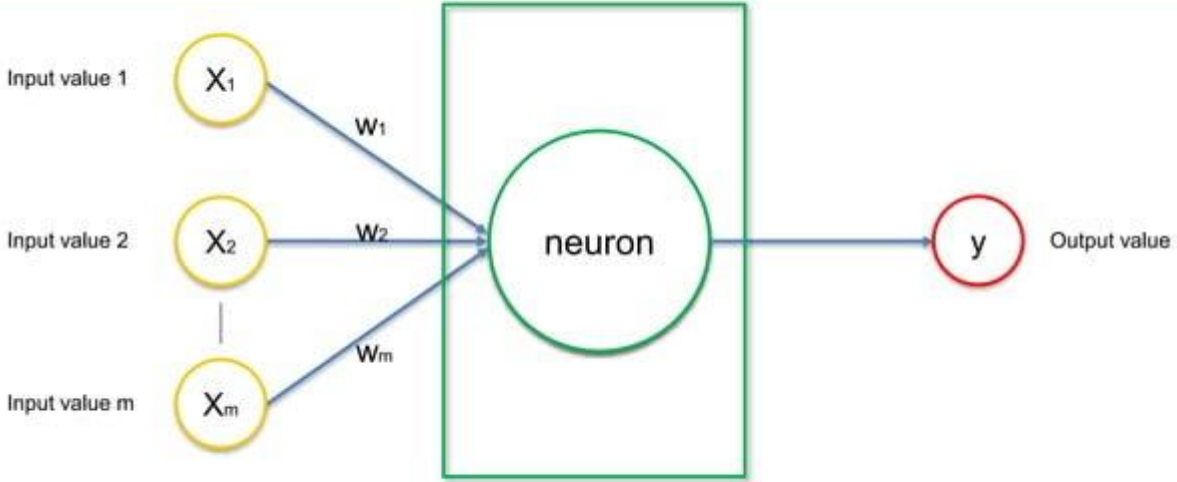

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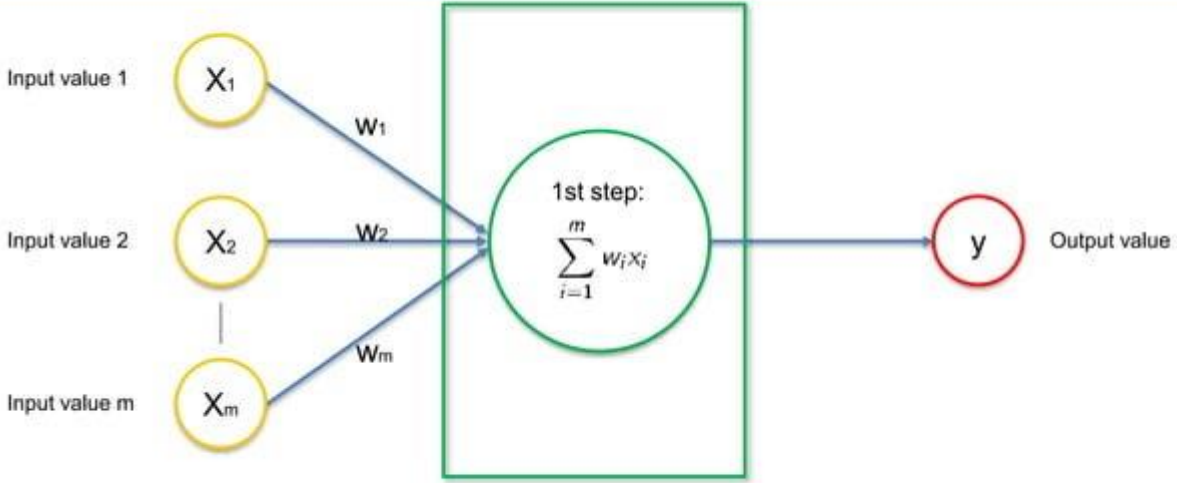

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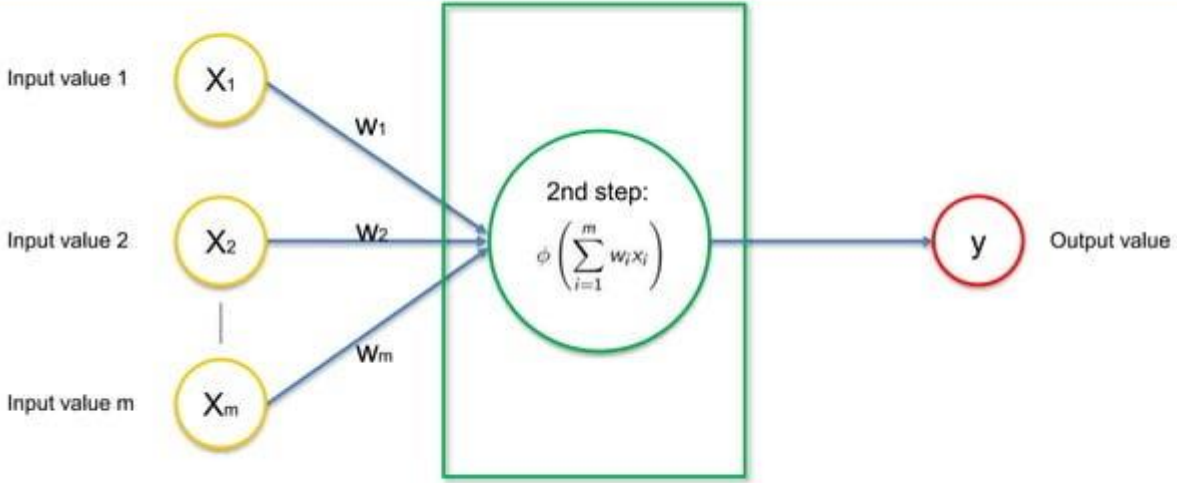

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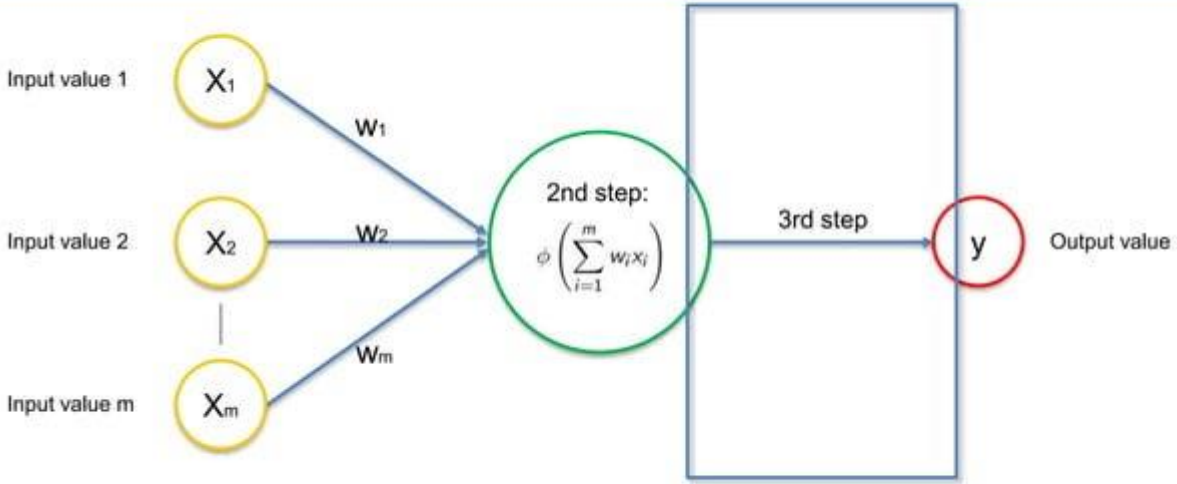
The Neuron



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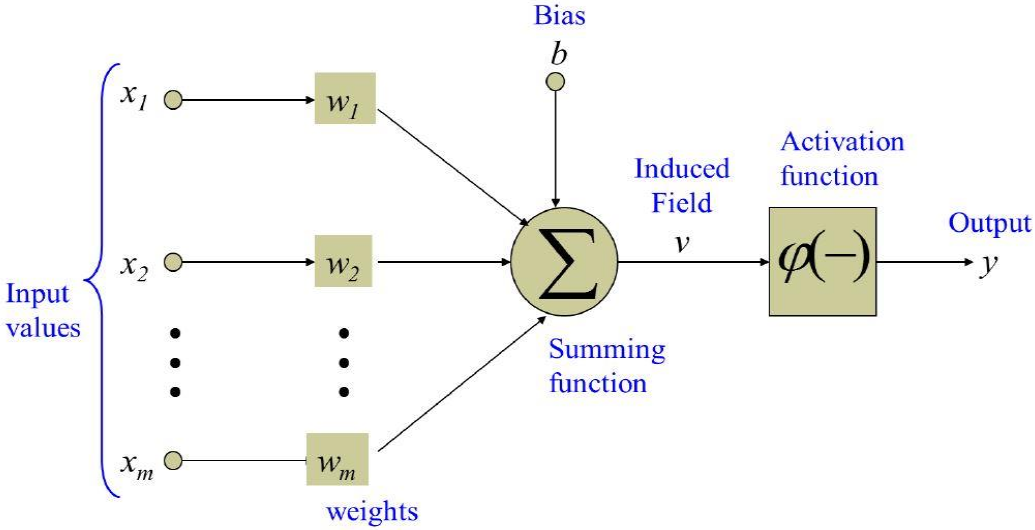
The basic unit of **Artificial Neural Networks**



The basic unit of **Artificial Neural Networks**

1. **A set of synapses** or connecting links, each of which is characterized by a weight or strength of its own. Specifically, a signal x_j at the input of synapse j connected to neuron k is multiplied by the synaptic weight w_j.

   Unlike a synapse in the brain, the synaptic weight of an artificial neuron may lie in a range that includes negative as well as positive values.

2. **An adder for summing** the input signals, weighted by the respective synapses of the neuron; the operations described here constitute a linear combiner.

3. **An activation function** for limiting the amplitude of the output of a neuron. The activation function is also referred to as a squashing function in that it squashes (limits) the permissible amplitude range of the output signal to some finite value.

Basic Structure of Neuron

- A **neuron** in an artificial neural network is a mathematical function that:
- Receives input(s)
- Multiplies them by **weights**, adds a **bias**,
- Passes the result through an **activation function** to produce output.
- **Mathematical Expression**:

$$y = \emptyset\left(\sum_{i=1}^{i=n} w_i\, x_i + b\right)$$

Where,

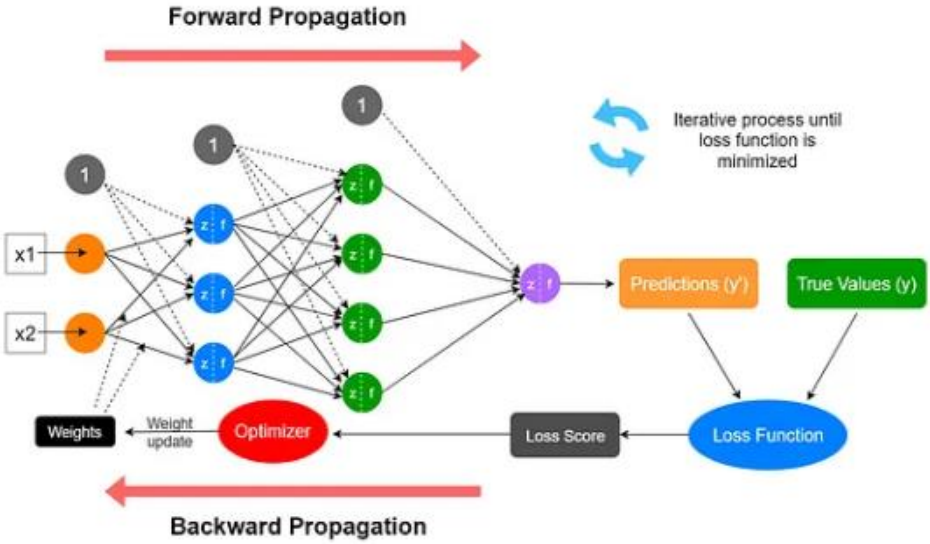
x_i = Input feature/signals

w_i = Synaptic weights of neuron

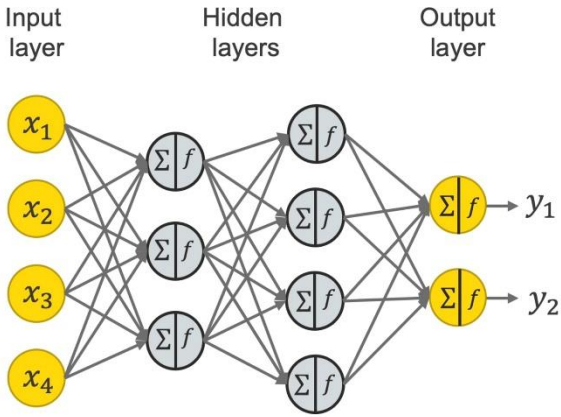
b = Bias

$\emptyset$ = Activation Function, y = Predicted Output

Artificial Neural Networks (Cont'd)

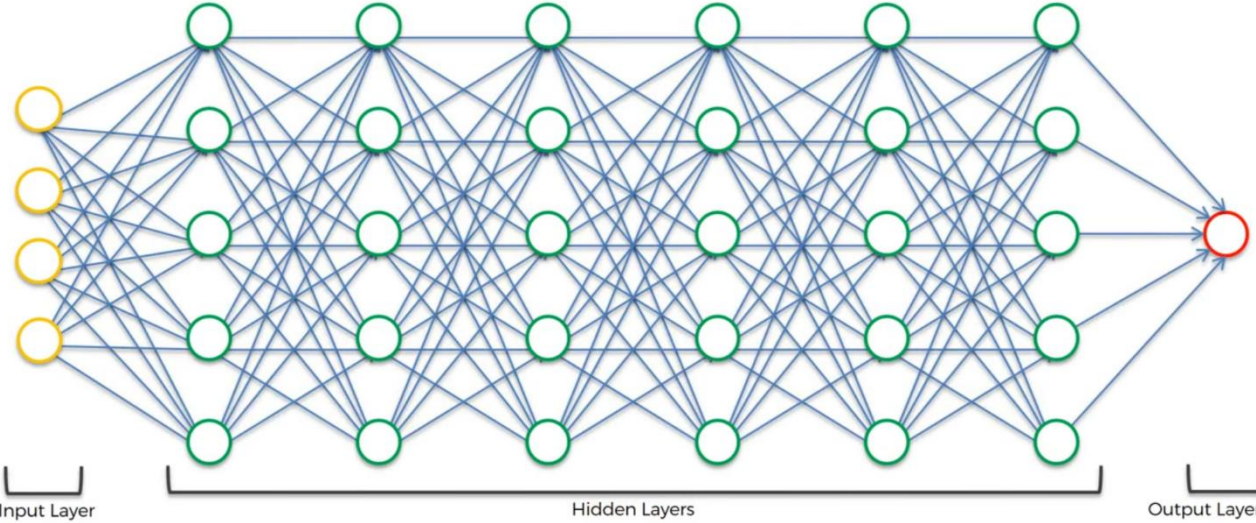


Architecture



- You need **one input node for each feature** of your input data.
- You need **one output node per class or output dimension**.

What is Deep Learning?



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Layers in Neural Networks

Neural networks consist of **layers** of neurons:

Input Layer

- The first layer which has input nodes
- Receives raw input data (e.g., feature vector).
- No processing or activation function is applied here.

Hidden Layer(s)

- One or more layers between input and output.
- Perform computations and extract patterns using weights and activation functions.
- Called **“hidden”** because their outputs are not directly observable.

Layers in Neural Networks (Cont'd)

Output Layer

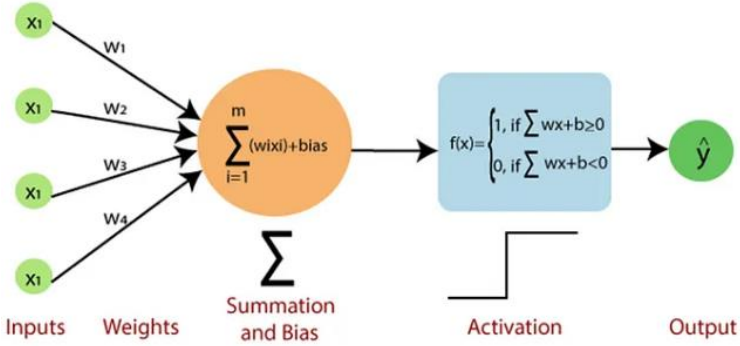
- Final layer which has output nodes that produces predictions.
- Activation depends on task:
  - **Sigmoid** for binary classification
  - **Softmax** for multiclass classification
  - **Linear** for regression

Different Types of Neural Networks

- Single Layer Perceptron
- Multilayer Perceptrons (MLPs)
- Feedforward Neural Networks (FNNs)
- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs)

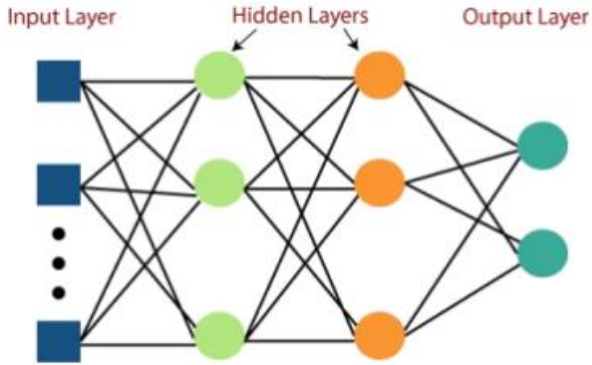
Single Layer Perceptron

A **Single Layer Perceptron** is the simplest form of a neural network that consists of **one input layer directly connected to one output layer**, without any hidden layers. It can only solve **linearly separable problems** using a basic decision boundary.



Multi Layer Perceptron (MLPs)

A **Multilayer Perceptron** is a type of feedforward artificial neural network that consists of **an input layer, one or more hidden layers, and an output layer**. It can solve both **linear and nonlinear problems** and is trained using **backpropagation**.



Applications of Neural Networks

1. Image Recognition

Neural networks can identify objects, faces, animals, handwriting, and patterns in images.

Example: A face unlock system in a smartphone uses neural networks to recognize the owner's face.

2. Speech Recognition

They can convert spoken language into text and understand voice commands.

Example: Google Assistant, Siri, and Alexa use neural networks to understand what users say.

3. Natural Language Processing

Neural networks help computers understand, generate, and translate human language.

Example: Google Translate uses neural networks to translate English into Bangla or other languages.

Applications of Neural Networks

4. Medical Diagnosis

They are used to analyze medical reports, X-rays, MRIs, and disease patterns.

Example: A neural network can help detect tumors from MRI scans or identify pneumonia from chest X-rays.

5. Self-Driving Cars

Neural networks help autonomous vehicles recognize roads, pedestrians, traffic lights, and nearby vehicles.

Example: A self-driving car uses neural networks to detect a red traffic signal and stop automatically.

6. Recommendation Systems

They analyze user behavior and suggest relevant products, movies, music, or videos.

Example: YouTube and Netflix use neural networks to recommend videos or movies based on your previous activity.

Applications of Neural Networks

7. Financial Forecasting

Neural networks can analyze market data and predict trends, risks, or fraud.

Example: Banks may use neural networks to detect unusual credit card transactions and identify fraud.

8. Handwriting Recognition

They can read handwritten characters, words, or numbers.

Example: Postal services use neural networks to read handwritten addresses on envelopes.

9. Chatbots and Virtual Assistants

Neural networks help chatbots understand questions and reply more naturally.

Example: Customer support chatbots on websites use neural networks to answer user queries.

10. Education

Neural networks are used in intelligent tutoring systems and automated grading.

Example: An online learning platform may use neural networks to recommend practice questions based on student performance.

The Activation Function

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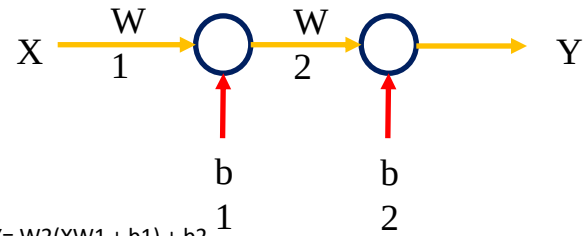
Activation Function

An **activation function** is a mathematical function used in a neural network **to decide whether a neuron should be "activated" or not** — meaning, whether the information that the neuron is processing is important enough to pass on to the next layer.

- Without activation functions, a neural network would simply be a linear system (like just adding and multiplying numbers).
- Activation functions introduce non-linearity, which allows the network to learn and model complex patterns and relationships in data.

Activation Function (Cont'd)

Lets consider a simple neural network without activation function:



$$Y = W2(XW1 + b1) + b2$$
$$= \underbrace{W2W1}_{W3}X + \underbrace{W2b1 + b2}_{b3}$$

$Y = W3X + b3$ → Which is nothing but a linear equation of X. So what is the use of multiple neuron here. What happened if we want to solve any non linear problem.

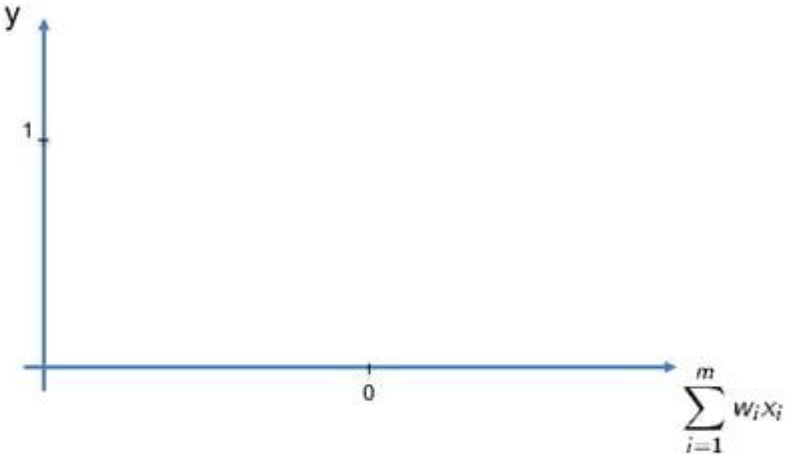
Activation Function (Cont'd)

Let's now insert a non-linear **activation function** f(·) like ReLU or sigmoid, **after the first layer**:

$$Y = ReLU[W2\{ReLU(W1X + b1)\} + b2]$$

Now we **cannot expand** into a single linear function of x, because Y is wrapped inside a non-linear function.

The Activation Function



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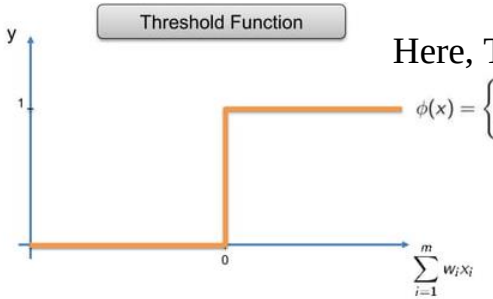
Types of Activation Function

- **Step / Threshold Function:**



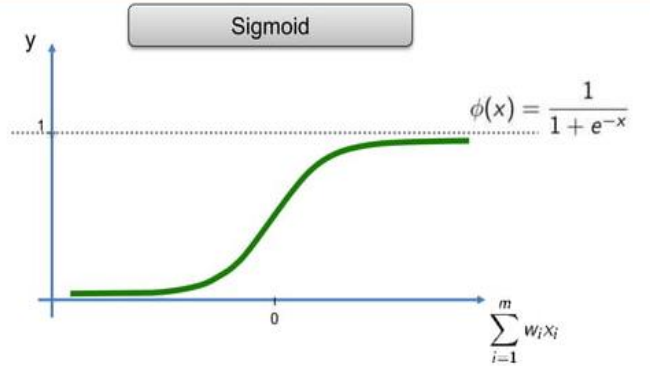
Here, Threshold = 0

$$\phi(x) = \begin{cases} 1 \text{ if } x \geq 0 \\ 0 \text{ if } x < 0 \end{cases}$$

- Threshold value can between 0 -1.
- The output of binary step function is either zero or 1. It is used in early binary classification (not used in deep learning).

Activation Function: Sigmoid

Sigmoid function is a mathematical function having a characteristic "S"-shaped curve or sigmoid curve. Often, It refers to the special case of the logistic function. It maps any real-valued number into a value between 0 and 1. Mainly used for binary classification (output layer).

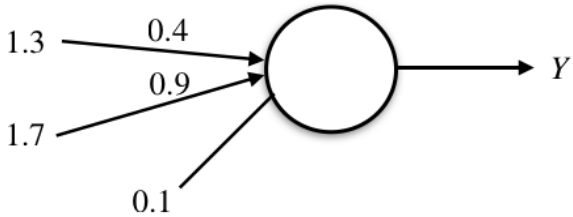


Activation Function: Sigmoid

The sigmoid function gives an 'S' shaped curve.
This curve has a finite limit of:
'0' as x approaches $-\infty$
'1' as x approaches $+\infty$
The output of sigmoid function when x=0 is 0.5
Thus, if the output is more than 0.5 , we can classify the outcome as 1 (or YES) and if it is less than 0.5 , we can classify it as 0(or NO) .
For example: If the output is 0.65, we can say in terms of probability as:
"There is a 65 percent chance that your favorite foot ball team is going to win today " .

Activation Function: Sigmoid

Find the output for the following neuron by sigmoid function-



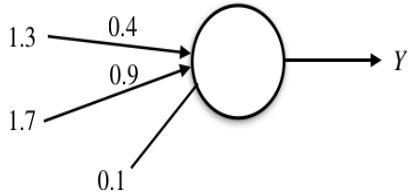
Here our x_0 and x_1 are 1.3 and 1.7. Weights are 0.4 , 0.9 and bias is 0.1

Activation Function: Sigmoid

$$\sum = w_0 x_0 + w_1 x_1 + b$$

$$\sum = 1.3 * 0.4 + 1.7 * 0.9 + 0.1$$

$$= 2.15$$



Now by sigmoid function we get-

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma(2.15) = \frac{1}{1 + e^{-2.15}}$$

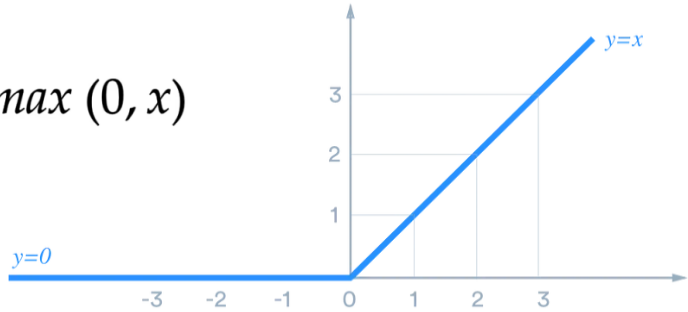
$$\sigma(2.15) = 0.89$$

Types of Activation Function (Cont'd)

- **ReLU (Rectified Linear Unit)**

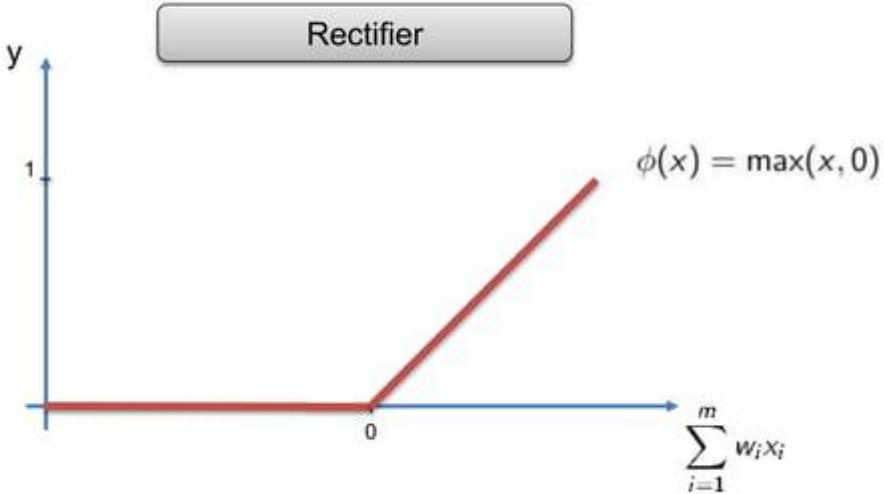
Mathematically ReLU can be represented as:

$$f(x) = max\ (0, x)$$



Basically used in the hidden layers of CNN or other deep neural networks. Its output is either zero (0) or as like the input.

The Activation Function



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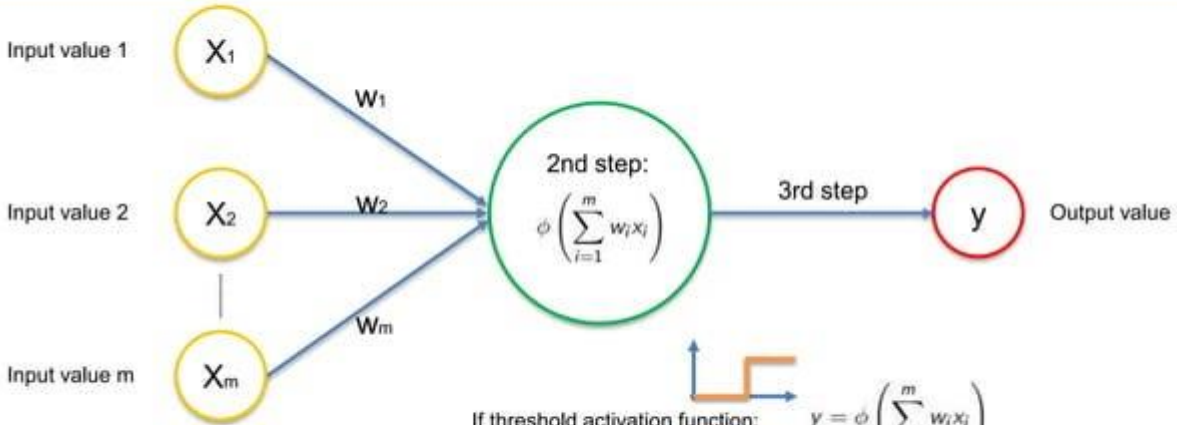
The Activation Function



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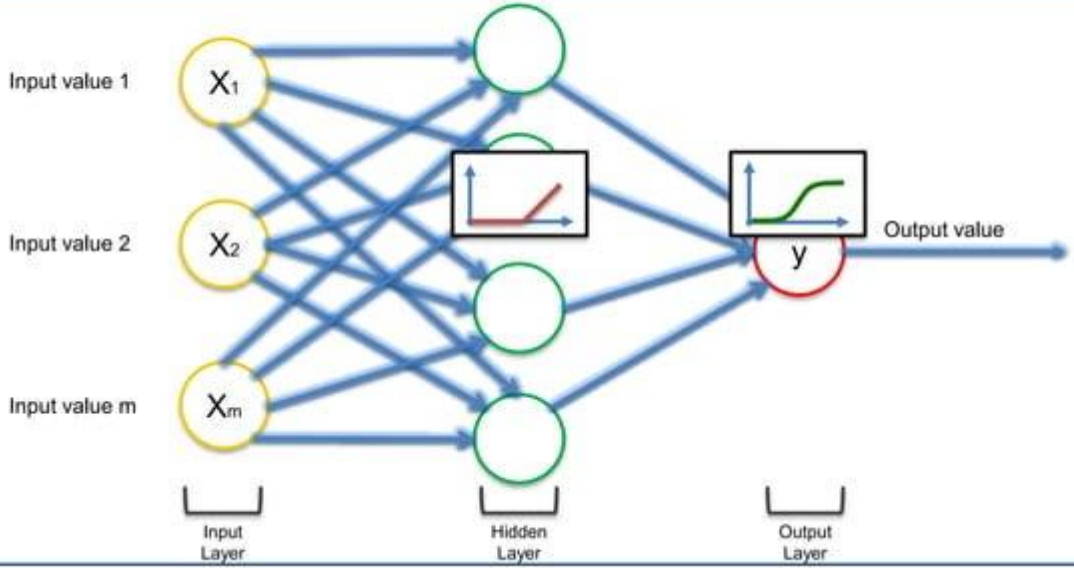
The Activation Function



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Bias

Bias is a trainable scalar value added to the weighted sum of inputs in a neuron before applying the activation function. It allows the model to better fit the data by enabling the activation function to shift horizontally, thereby adjusting the decision boundary.

Lets Imagine a Neuron Without Bias

- A neuron calculates: z = w·x, Then passes z through an **activation function**, like a step function and mathematically represented by,

$$\text{activation}(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Bias (Cont'd)

- So with, $Z = w \cdot x$, If $X = 0$, then $Z = 0$, and the neuron activates. But **we have no control** over when it activates—it always activates when $Z \geq 0$.

Now Add Bias:

- $Z = w \cdot x + b$, Now we can **shift** when the neuron activates:
- If $b = -1$, the neuron activates only when $w \cdot x \geq -1$
- If $b = +1$, the neuron activates even when $w \cdot x \geq 1$

Bias (Cont'd)

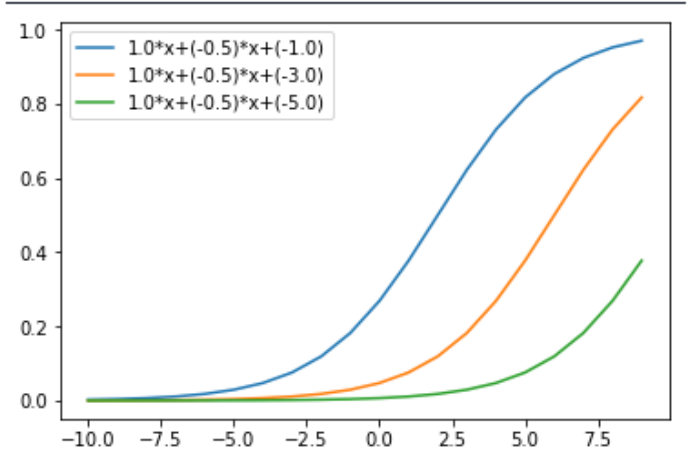
📊 Example With Step Function:

Let's say $w = 1$, and use:

$$z = x + b$$

x	b = 0 (No Bias)	b = -2	b = +2
-1	0 (off)	-3 (off)	+1 (on)
0	0 (on)	-2 (off)	+2 (on)
+1	+1 (on)	-1 (off)	+3 (on)

Bias (Cont'd)


1.0*x+(-0.5)*x+(-1.0)
1.0*x+(-0.5)*x+(-3.0)
1.0*x+(-0.5)*x+(-5.0)
1.0
0.8
0.6
0.4
0.2
0.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5

Loss Function

A **loss function** is a mathematical function that measures the **difference between the predicted output and the actual (true) output**. It helps a model learn by **penalizing incorrect predictions**—the larger the error, the larger the loss. During training, optimization algorithms (like gradient descent) minimize this loss function to improve model performance. For Example,

Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

How do NNs Learn?

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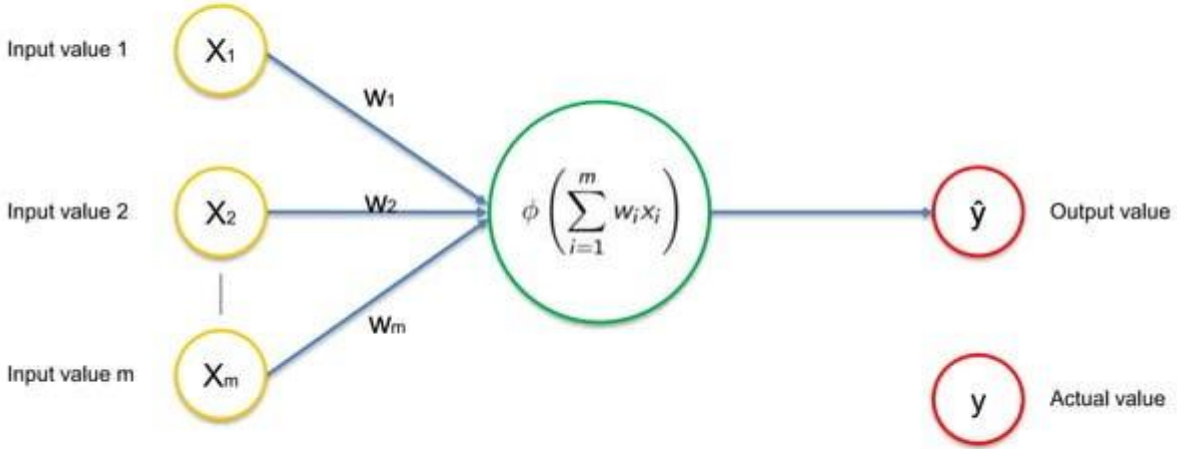
How do Neural Networks learn?



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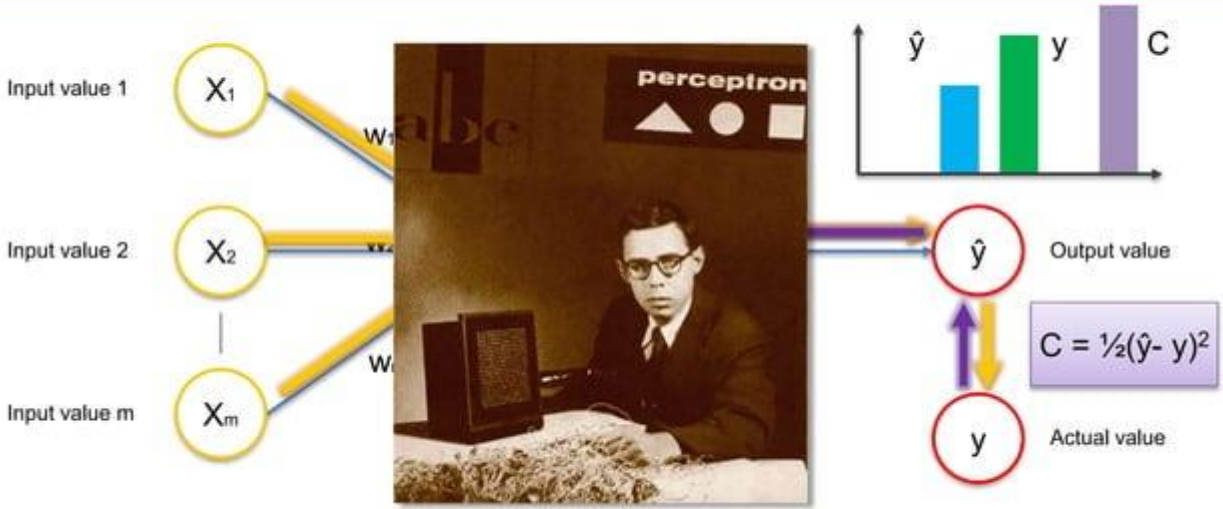
How do Neural Networks learn?



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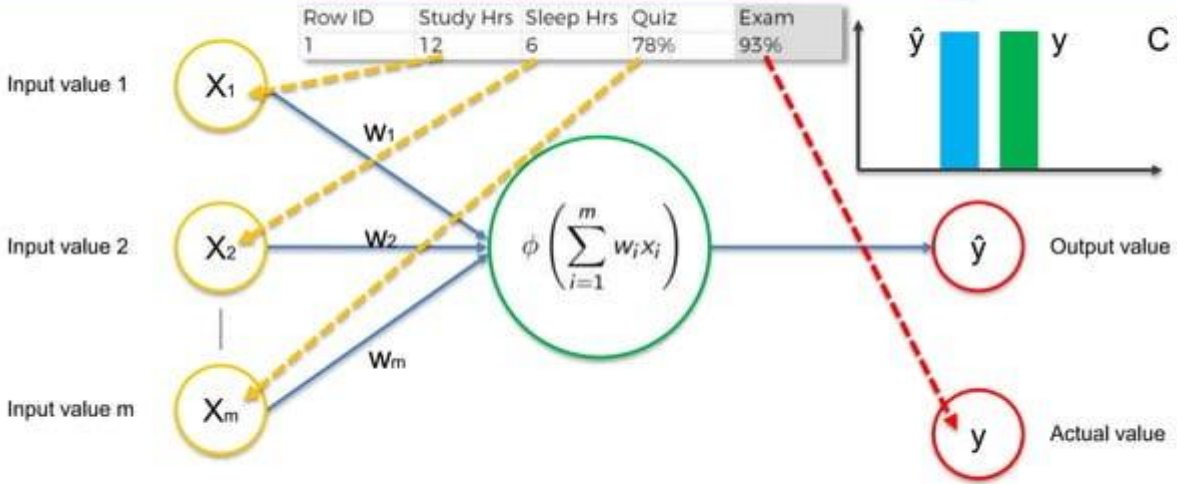

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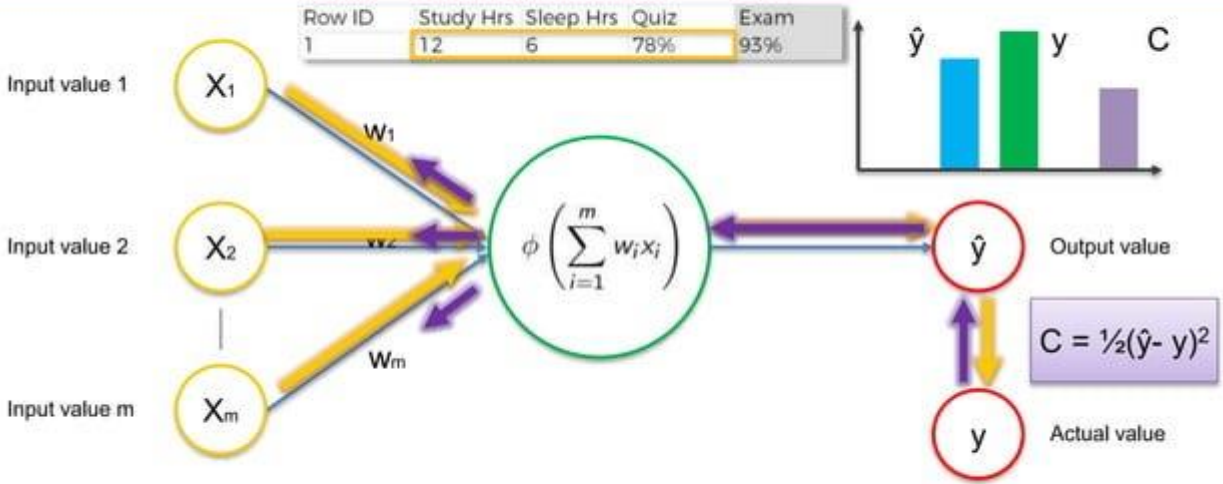

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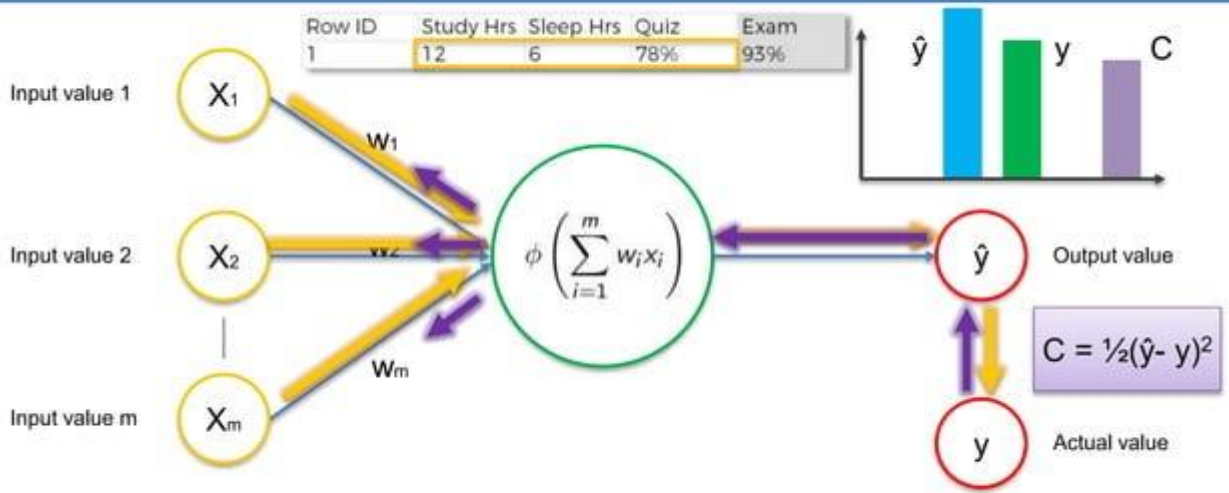

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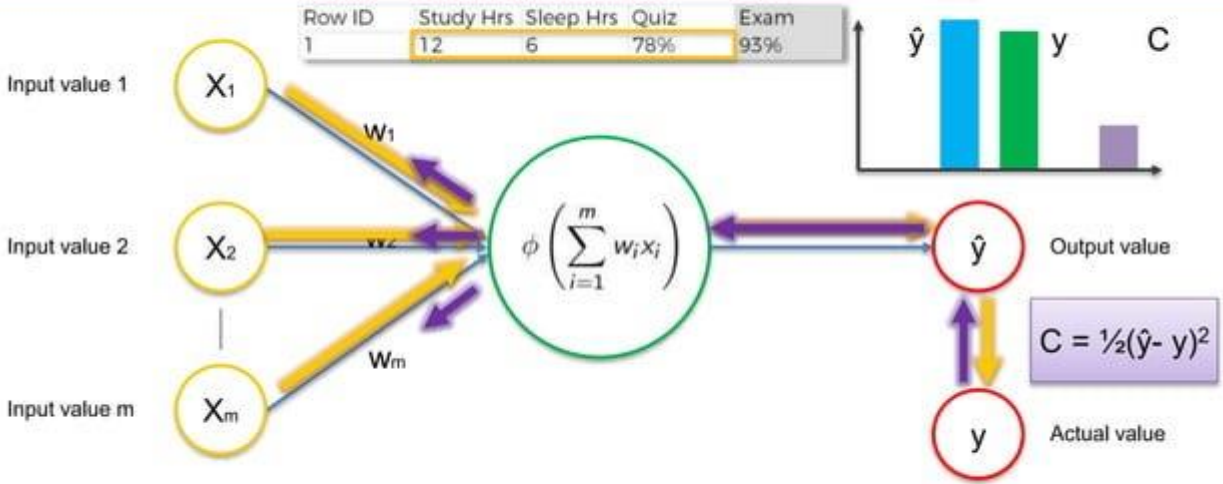

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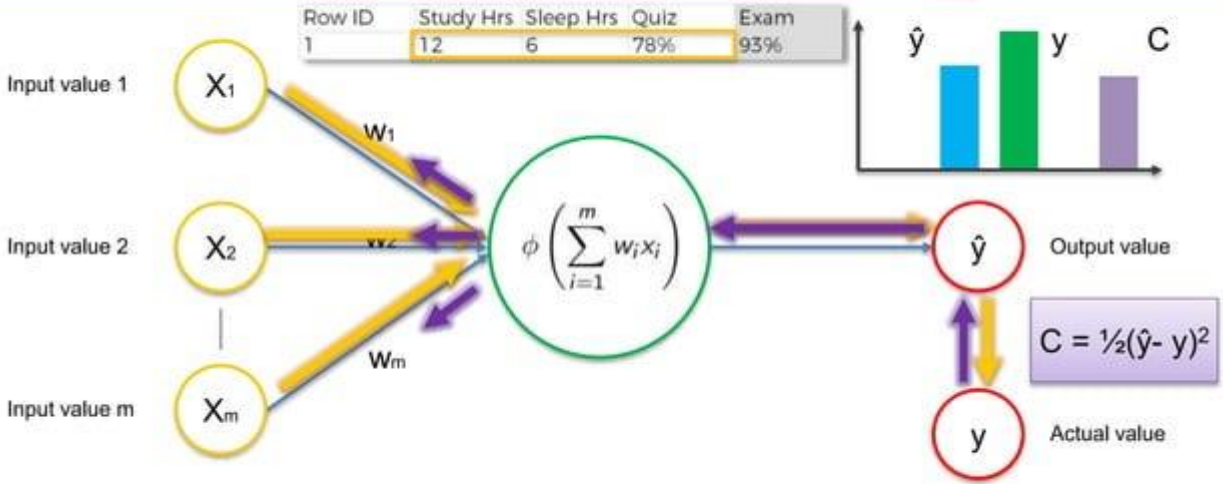

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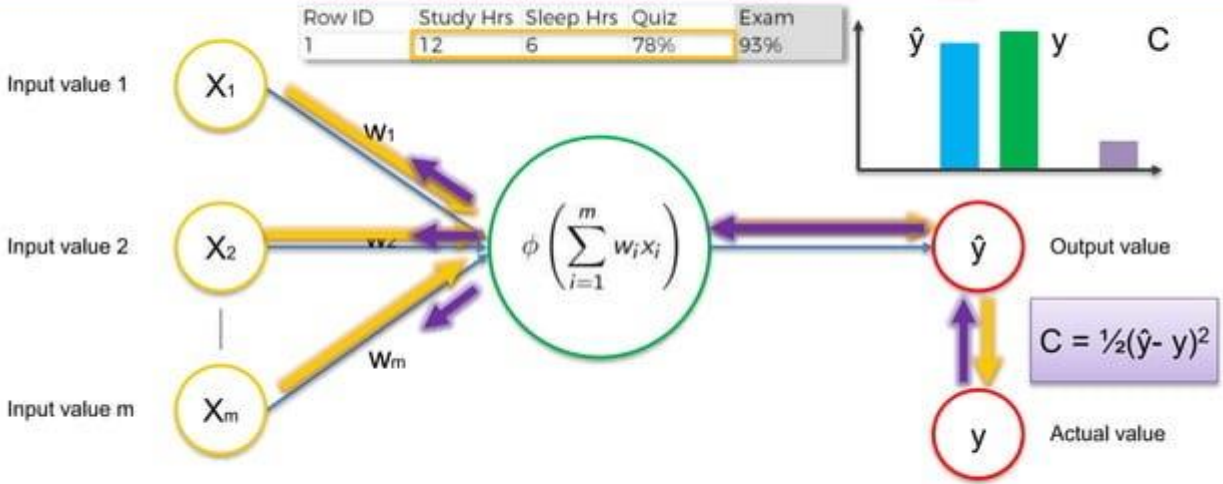
How do Neural Networks learn?



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How do Neural Networks learn?

Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%

Input value 1 X_1 w_1

Input value 2 X_2 w_2

Input value m X_m w_m

$$\phi\left(\sum_{i=1}^{m} w_i x_i\right)$$

$\hat{y}$ Output value

$$C = \tfrac{1}{2}(\hat{y} - y)^2$$

y Actual value

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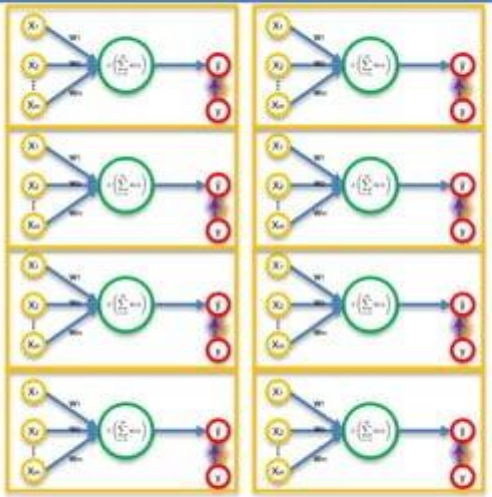
How do Neural Networks learn?



Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%
2	22	6.5	24%	68%
3	115	4	100%	95%
4	31	9	67%	75%
5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

$$C = \sum \tfrac{1}{2}(\hat{y} - y)^2$$

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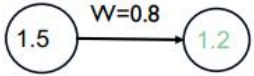
Train a Neural Network

We want to train a neural network on a dataset.

How?

1. Forward propagation
2 . Back propagation

Input	Actual output	Predicted output
1.5	0.5	1.2



$$\hat{y} = g\left(\sum_{i=1}^{m} x_i w_i\right)$$

What is training?

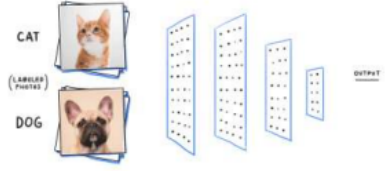
The model (NN) learns the relationship between the input and output data (supervised learning)

Adjusting the weights and biases; so that the error between the predicted output and actual output is a minimum.

Train a Neural Network

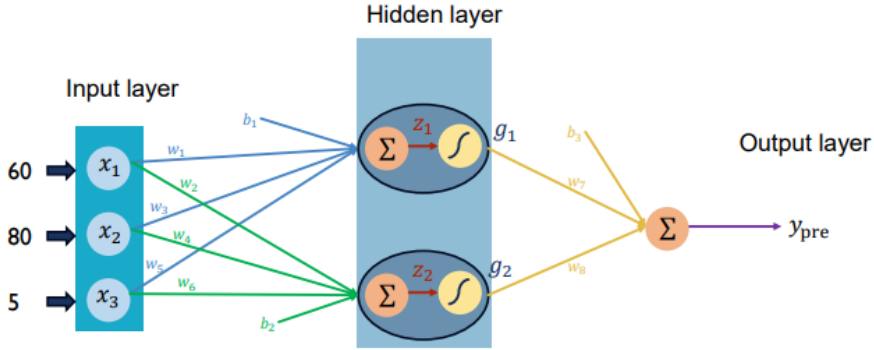
	Student	X1 Science grade [%]	X2 Chemistry grade [%]	X3 # hours studied [hrs.]	Y_{act} Final grade [%]
Train set	1	60	80	5	82
	2	70	75	7	94
	3	50	55	10	45
Test set	4	40	56	7	43

In classification problem there the output is a class such as dog or cat or male or female



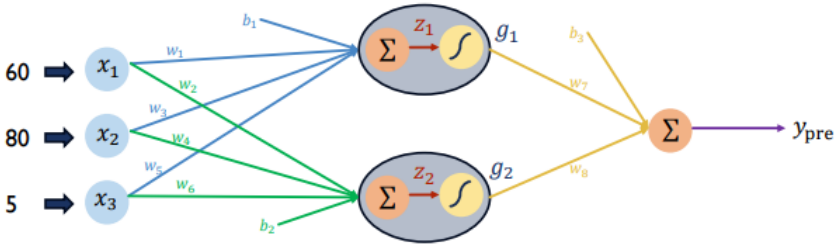
This is a regression problem because the output is a continuous features

Forward propagation

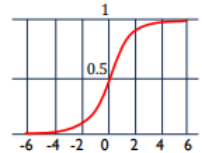


Student	X1 Science grade [%]	X2 Chemistry grade [%]	X3 # hours studied [hrs.]	Y_{act} Final grade [%]
1	60	80	5	82
2	70	75	7	94
3	50	55	10	45
4	40	56	7	43

Forward propagation



$$g_1 = \frac{1}{1 + e^{-z_1}}$$



$$z_1 = w_1 * x_1 + w_3 * x_2 + w_5 * x_3 + b_1$$
$$= 0.1 * 60 + 0.1 * 80 + 0.1 * 5 + (-15)$$
$$= -0.5$$

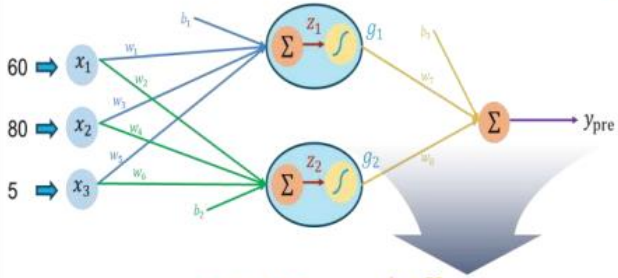
$$g_1 = \frac{1}{1 + e^{-(-0.5)}} = 0.37$$

Student	X1 Science grade [%]	X2 Chemistry grade [%]	X3 # hours studied [hrs.]	Y_{act} Final grade [%]
1	60	80	5	82
2	70	75	7	94
3	50	55	10	45
4	40	56	7	43

$$z_2 = w_2 * x_1 + w_4 * x_2 + w_6 * x_3 + b_2$$
$$= 0.15 * 60 + 0.05 * 80 + (-0.2) * 5 + (-3)$$
$$= -3$$

$$g_2 = 0.047$$

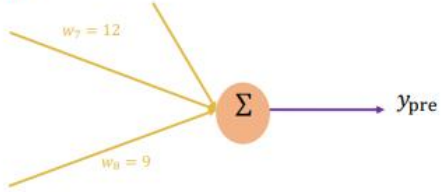
Forward propagation



Student	X1 Science grade [%]	X2 Chemistry grade [%]	X3 # hours studied [hrs.]	y_{act} Final grade [%]
1	60	80	5	82
2	70	75	7	94
3	50	55	10	45
4	40	56	7	43

$g_1 = 0.37$

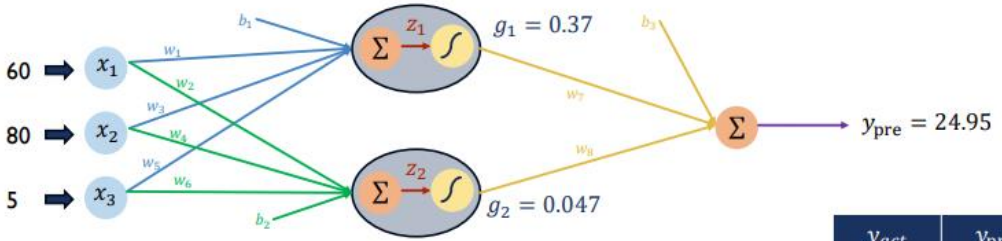
$b_3 = 20$



$g_2 = 0.047$

$$y_{pre} = w_7 * g_1 + w_8 * g_2 + b_3$$
$$= 12 * 0.37 + 9 * 0.047 + 20$$
$$= 24.95$$

Forward propagation



y_{act}	y_{pre}
82	24.95

Student	X1 Science grade [%]	X2 Chemistry grade [%]	X3 # hours studied [hrs.]	Y_{act} Final grade [%]
1	60	80	5	82
2	70	75	7	94
3	50	55	10	45
4	40	56	7	43

Adjusting the weights and biases; so that the error between the predicted output and actual output is a minimum.

Learning Rate

Learning rate(**η**) is a hyperparameter that governs how much a machine learning model adjusts its parameters at each step of its optimization algorithm. The learning rate can determine whether a model delivers optimal performance or fails to learn during the training process.